

Tushar Jayendra Mhatre

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PROFESSIONAL SUMMARY

Software Developer Full-stack software developer with 3+ years building web applications, REST APIs, and AI-driven pipelines. At Aether Loop, built a React/TypeScript frontend, designed FastAPI REST services, and architected multi-agent AI systems using LangChain and CrewAI. Currently completing a Master's in Computer Science at OU with a 3.97 GPA, with research experience building ML-driven tools for neuroimaging data analysis as a GRA. Comfortable with Docker, GCP, and working in interdisciplinary research environments where the work has real-world impact.

EDUCATION

University of Oklahoma

Master of Science in Data Science & Analytics(GPA: 3.97)

Norman,OK

May 2025

Savitribai Phule Pune University

Bachelor of Engineering in Electronics and Telecommunications(GPA: 7.6/10)

Pune, MH,India

May 2020

PROFESSIONAL EXPERIENCE

The Aether Loop

AI Engineering Intern

Remote

September 2025-present

- Architected a multi-agent AI recommendation engine using CrewAI and LangChain, implementing agent orchestration, tool use, and structured clinical insight extraction from 40+ user biomarkers.
- Conducted iterative model evaluations and human-in-the-loop validation cycles, improving model performance from average Protocol score of 67% to 90%
- Engineered and maintained a scalable FastAPI backend deployed on GCP Cloud Run with Docker and CI/CD pipelines, sustaining 99% uptime with continuous model monitoring and improvement.
- Built and enhanced complex React/TypeScript frontend features using Vite, including constructing a full admin portal derived from the main application to support internal user management and operational workflows.
- Integrated Stripe into the platform to support multi-tier subscription billing, implementing webhook handling for payment events and building out subscription lifecycle management across multiple timeframes.

Optimal Living Systems

Data Architect Intern

Norman, OK

July 2025 - September 2025

- Designed and deployed the initial ETL infrastructure in Airflow,OCR(Docling) and OpenAI API, to transform 1,000+ PDFs into structured QnA datasets to kick off the organization's first LLM fine-tuning pilot.
- Built reusable Python (BeautifulSoup, Requests) scraping scripts for the research team, reducing their manual sourcing by 3-4 hours daily, leading to completion of data targets two weeks ahead of deadline.

University Of Oklahoma

Graduate Research Assistant

Norman, OK

February 2024 - May 2025

- Built executable EEG/MRI BIDS transformation pipelines in MATLAB, reducing manual structuring time from 5+ hours per dataset to single-click execution which enabled background preprocessing while researchers prioritized core research work.
- Developed WSL-based Bash scripts integrating Windows/MATLAB workflows with Linux tools (FSL, AFNI), reducing preprocessing runtime by 40% and accelerating experiments turnaround through parallel job scheduling.
- Led a year-long neuroimaging research study applying clustering algorithms across EEG datasets to identify and analyze Cortical Activity Patterns (CAPs), validating findings through statistics and co-authored a manuscript under review at Brain Topography.

Tata Consultancy Services

Software Engineer

Pune, MH,India

February 2021 - July 2023

- Developed 9 RPA bots using Automation 360, VBA, and SQL over 6 months, reducing manual workload by 20-25 hours weekly and driving \$100K+ annual cost savings for enterprise clients through SAP and mainframe automation.
- Migrated 13 workflows from Automation Anywhere v11 to Automation 360 in 2 months, ensuring zero downtime and fewer deployment defects while transitioning the team to a more scalable and maintainable automation platform.
- Created a Python REST API integrated with Automation Anywhere bots to retrieve product toxicity ratings from a third party service, reducing data lookup time from minutes to seconds and accelerating workflow completion time.

- Deployed a classification model via a Flask API integrated into an RPA workflow, which predicted invoice discrepancy risk across high-volume finance operations and processed 1K+ predictions daily to support faster, data-driven reconciliation decisions.
- Built a Power Automate flow to auto-sync open issue statuses from SharePoint into Jira on a daily schedule, eliminating manual cross-platform updates for the support team and cutting ticket update lag from 24+ hours to near real-time, saving 5–6 hours weekly

PROJECTS

Flappy Kiro: Cloud-Native Kubernetes Gaming Platform

- Architected a Kubernetes-based cloud gaming platform on AWS EKS with ALB/ECR-backed microservices and end-to-end observability, sustaining 99.9% uptime and sub-100ms responses, by containerizing Docker builds and deploying 11 Edge Delta monitors.

NLP Job Posting Fraud Detection Classification System

- Built an NLP-driven fake-job detection portal achieving 96% accuracy and 85% minority-class true positive rate despite severe class imbalance, improving fraud identification for high-risk postings, by training and tuning scikit-learn models (XGBoost, SVM, Random Forest, Logistic Regression).

LangGraph Data Analysis Agent with Secure Python Sandbox

- Engineered an AI data-analysis agent that converts natural-language queries into vetted, sandboxed Python transformations, executing analyses in <2 seconds median end-to-end while blocking unsafe code paths, by orchestrating LangGraph workflows with React + FastAPI.

TECHNICAL SKILLS

Programming Languages: Python, R, SQL, C/C++, MATLAB, Go

Python Libraries: NumPy, Pandas, matplotlib, Scikit-learn, Tensorflow, XGboost, Pytorch, Seaborn, imblearn, Flask, Stripe

Frontend: React, TypeScript, Vite, HTML/CSS

Devops and Cloud: Microsoft Azure, Docker, Google Cloud Platform, Vertex AI, CI/CD, Git, BigQuery, Supabase

Generative AI: LangChain, Ollama, Langraph, Crew AI, Hugging Face, RAG, Unsloth AI

Certifications: Google Data Analytics Professional Certification, Advanced C++ certification (Udemy)

AI Tools: Cursor, Claude Code, Augment, Github Copilot, Gamma, ElevenLabs